LESSON 5.10

Volume of Rectangular Prisms

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the volume of a rectangular prism, including ones with fractional edge lengths, using base area × height.

WHY THIS MATTERS Your class is building a time capsule to bury on school grounds. The capsule is a rectangular prism that is 2 feet long, 1.5 feet wide, and 1 foot tall. You need to figure out how much stuff can fit inside — that means calculating the volume!

REMEMBER

Volume of Rectangular Prisms

Volume of Rectangular Prisms with Fractional Edges. Formula: V = l × w × h = B × h Volume is the space inside a rectangular prism. There are two ways to write the same rule. You can multiply all three edges, V = length × width × height. Or you can find the base area first, B = length × width, and then stack that base up the height: V = B × h. Both give the same number, and both work when an edge is a fraction or a decimal.

1. Convert mixed numbers to improper fractions before multiplying.
2. Multiply numerators across, multiply denominators across: V = (l · w · h).
3. Simplify the fraction and label with cubic units

Words: **Volume *(Volumen)*** **Rectangular prism *(Prisma rectangular)*** **Cubic units *(Unidades cúbicas)*** **Dimensions *(Dimensiones)*** **Base area *(Área de la base)***

1 **Circle the letter of the best answer. Show how you know.**

A cube has an edge length of 5 cm. What is its volume?

(A) 15 cm³ (B) 25 cm³

(C) 75 cm³ (D) 125 cm³

HINT A cube's edges are all equal, so its length, width, and height are each 5 cm.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

2 **Circle the letter of the best answer. Show how you know.**

A regular hexagon splits into 6 identical triangles, each with an area of 8 square cm. What is the area of the hexagon?

(A) 8 square cm
(B) 14 square cm
(C) 48 square cm
(D) 48 cm

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

A regular pentagon is decomposed into triangles from its centre. How many triangles are there?

(A) 3
(B) 5
(C) 6
(D) 10

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A regular octagon has an area of 96 square inches and splits into 8 identical triangles. What is the area of one triangle?

(A) 8 square inches

(B) 12 square inches

(C) 24 square inches

(D) 768 square inches

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

5 **Circle the letter of the best answer. Show how you know.**

Why does decomposing a regular polygon into triangles help you find its area?

(A) Triangles have no area

(B) Every triangle's area is easy to find, and the parts add to the whole

(C) It turns the polygon into a rectangle

(D) It makes the perimeter shorter

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

An aquarium store sells fish tanks. A small tank is 24 inches long, 12 inches wide, and 16 inches tall. One gallon of water takes up about 231 cubic inches. What is the tank's volume?

(A) 52 in³
(B) 576 in³
(C) 4,608 in³
(D) 4,806 in³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

EXPLAIN YOUR THINKING

Capsules A and B have the same volume (3 ft³) but different dimensions. Why? Which design would you choose and why?

SENTENCE STARTER Capsules A and B both have V = ___ because ___ × ___ × ___ = ___ × ___ × ___. I'd choose ___ because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each box's dimensions to its correct volume.

_____ 2 × 3 × 4	(A) 24 cubic in
_____ 5 × 2 × 2	(B) 20 cubic in
_____ 3 × 3 × 3	(C) 27 cubic in
_____ 6 × 1 × 5	(D) 30 cubic in

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

An aquarium store sells fish tanks. A small tank is 24 inches long, 12 inches wide, and 16 inches tall. One gallon of water takes up about 231 cubic inches. About how many gallons does the tank hold?

(A) about 2 (B) about 20

(C) about 200 (D) about 231

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

An aquarium store sells fish tanks. A small tank is 24 inches long, 12 inches wide, and 16 inches tall. One gallon of water takes up about 231 cubic inches. Why divide by 231 instead of multiplying?

(A) Because dividing always makes numbers smaller

(B) Because you are finding how many 231-in³ gallons fit inside the tank

(C) Because 231 is bigger than 24

(D) Because gallons are bigger than inches

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A time-capsule box measures 6 in long, 4 in wide, and 3 in tall. What is its volume?

(A) 13 cubic in

(B) 24 cubic in

(C) 72 cubic in

(D) 72 square in

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

5 **Circle the letter of the best answer. Show how you know.**

Two storage boxes are ready to pack. Box A is 5 × 5 × 2 inches and Box B is 4 × 4 × 3 inches. Which box holds more?

(A) Box B, because 48 cubic in is more than 50 cubic in

(B) Box A, because 50 cubic in is more than 48 cubic in

(C) They hold the same amount

(D) Box B, because it is taller

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A keepsake box is 4 in long, 3 in wide, and 1½ in tall. What is its volume?

(A) 8.5 cubic in

(B) 12 cubic in

(C) 18 cubic in

(D) 36 cubic in

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

EXPLAIN YOUR THINKING

Capsules A and B have the same volume (3 ft³) but different dimensions. Why? Which design would you choose and why?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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